Fermi's golden rule

Find the transition rate $R_{1\to2}$ for the perturbing Hamiltonian

$$H_1(\mathbf{r},t) = 2V(\mathbf{r})\cos(\omega t + \phi)$$

§1

Let $\Psi(\mathbf{r},t)$ be the following linear combination of the two eigenstates.

$$\Psi(\mathbf{r},t) = c_1(t)\psi_1(\mathbf{r})\exp(-i\omega_1 t) + c_2(t)\psi_2(\mathbf{r})\exp(-i\omega_2 t)$$

We need to solve for $c_2(t)$ to find the transition rate $R_{1\to2}$.

Start with the Schrodinger equation

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = H_0(\mathbf{r})\Psi(\mathbf{r},t) + H_1(\mathbf{r},t)\Psi(\mathbf{r},t)$$

where

$$H_0(\mathbf{r})\Psi(\mathbf{r},t) = E_1 c_1(t)\psi_1(\mathbf{r})\exp(-i\omega_1 t) + E_2 c_2(t)\psi_2(\mathbf{r})\exp(-i\omega_2 t)$$

Evaluate the time derivative.

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = i\hbar\frac{\partial c_1(t)}{\partial t}\psi_1(\mathbf{r})\exp(-i\omega_1 t) + \boxed{\hbar\omega_1 c_1(t)\psi_1(\mathbf{r})\exp(-i\omega_1 t)}\\
+ i\hbar\frac{\partial c_2(t)}{\partial t}\psi_2(\mathbf{r})\exp(-i\omega_2 t) + \boxed{\hbar\omega_2 c_2(t)\psi_2(\mathbf{r})\exp(-i\omega_2 t)}\\
= H_0(\mathbf{r})\Psi(\mathbf{r},t) + H_1(\mathbf{r},t)\Psi(\mathbf{r},t)
\end{aligned}$$

Noting that $\hbar\omega_1 = E_1$ and $\hbar\omega_2 = E_2$, the boxed terms cancel with $H_0(\mathbf{r})\Psi(\mathbf{r},t)$ leaving

$$i\hbar\frac{\partial c_1(t)}{\partial t}\psi_1(\mathbf{r})\exp(-i\omega_1 t) + i\hbar\frac{\partial c_2(t)}{\partial t}\psi_2(\mathbf{r})\exp(-i\omega_2 t) = H_1(\mathbf{r},t)\Psi(\mathbf{r},t)$$

Substitute for $H_1(\mathbf{r})$ and $\Psi(\mathbf{r},t)$.

$$\begin{aligned}
i\hbar\frac{\partial c_1(t)}{\partial t}\psi_1(\mathbf{r})\exp(-i\omega_1 t) + i\hbar\frac{\partial c_2(t)}{\partial t}\psi_2(\mathbf{r})\exp(-i\omega_2 t)\\
= 2V(\mathbf{r})\cos(\omega t + \phi)\Big[c_1(t)\psi_1(\mathbf{r})\exp(-i\omega_1 t) + c_2(t)\psi_2(\mathbf{r})\exp(-i\omega_2 t)\Big]
\end{aligned}$$

Take the inner product of $\psi_2^*(\mathbf{r})$ with both sides to obtain

$$i\hbar\frac{\partial c_2(t)}{\partial t}\exp(-i\omega_2 t) = 2\cos(\omega t + \phi)\Big[c_1(t)M_{21}\exp(-i\omega_1 t) + c_2(t)M_{22}\exp(-i\omega_2 t)\Big]$$

where M_{21} and M_{22} are the matrix elements

$$M_{21} = \int \psi_2^*(\mathbf{r})V(\mathbf{r})\psi_1(\mathbf{r})\,d\mathbf{r}$$

$$M_{22} = \int \psi_2^*(\mathbf{r})V(\mathbf{r})\psi_2(\mathbf{r})\,d\mathbf{r}$$

Multiply both sides by $\exp(i\omega_2 t)$.

$$i\hbar\frac{\partial c_2(t)}{\partial t} = 2\cos(\omega t + \phi)\Big[c_1(t)M_{21}\exp[i(\omega_2 - \omega_1)t] + c_2(t)M_{22}\Big]$$

Let the initial state be $\Psi(\mathbf{r}, 0) = \psi_1(\mathbf{r})$ hence the initial conditions are

$$c_1(0) = 1, \quad c_2(0) = 0$$

For time t near the origin use $c_1(t) \approx 1$ and $c_2(t) \approx 0$ to obtain

$$i\hbar\frac{\partial c_2(t)}{\partial t} = 2\cos(\omega t + \phi)M_{21}\exp[i(\omega_2 - \omega_1)t]$$

Solve for $c_2(t)$ by integrating.

$$c_2(t) = \frac{2M_{21}}{i\hbar}\int_0^t \cos(\omega t' + \phi)\exp[i(\omega_2 - \omega_1)t']\,dt'$$

In exponential form

$$c_2(t) = \frac{M_{21}}{i\hbar}\int_0^t \exp(i\omega t' + \phi)\exp[i(\omega_2 - \omega_1)t']\,dt' + \frac{M_{21}}{i\hbar}\int_0^t \exp(-i\omega t' - \phi)\exp[i(\omega_2 - \omega_1)t']\,dt'$$

The solution is

$$c_2(t) = -\frac{M_{21}}{\hbar}\,\frac{\exp[i(\omega_2 - \omega_1 + \omega)t] - 1}{\omega_2 - \omega_1 + \omega}\,\exp(i\phi) - \frac{M_{21}}{\hbar}\,\frac{\exp[i(\omega_2 - \omega_1 - \omega)t] - 1}{\omega_2 - \omega_1 - \omega}\,\exp(-i\phi) \qquad (1)$$

For ω such that $\omega \approx \omega_2 - \omega_1$ the second term dominates so discard the first term.

$$c_2(t) = -\frac{M_{21}}{\hbar}\,\frac{\exp[i(\omega_2 - \omega_1 - \omega)t] - 1}{\omega_2 - \omega_1 - \omega}\,\exp(-i\phi)$$

By the identity

$$\exp(ix) - 1 = 2i\sin\left(\frac{x}{2}\right)\exp\left(\frac{ix}{2}\right)$$

we have

$$c_2(t) = -\frac{2iM_{21}}{\hbar}\,\frac{\sin\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right]\exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right]}{\omega_2 - \omega_1 - \omega}\,\exp(-i\phi)$$

Multiply numerator and denominator by $t/2$.

$$c_2(t) = -\frac{itM_{21}}{\hbar}\,\frac{\sin\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right]\exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right]}{\frac{1}{2}(\omega_2 - \omega_1 - \omega)t}\,\exp(-i\phi)$$

Substitute $\operatorname{sinc}(x)$ for $\sin(x)/x$.

$$c_2(t) = -\frac{itM_{21}}{\hbar}\operatorname{sinc}\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right]\exp\left[\frac{1}{2}i(\omega_2 - \omega_1 - \omega)t\right]\exp(-i\phi) \qquad (2)$$

§2

Transition density P is squared magnitude of $c_2(t)$.

$$P = |c_2(t)|^2 = \frac{t^2}{\hbar^2}\,|M_{21}|^2\,\mathrm{sinc}^2\left[\frac{1}{2}(\omega_2 - \omega_1 - \omega)t\right] \tag{3}$$

Total density P_{tot} is

$$P_{tot} = \frac{t^2}{\hbar^2}|M_{21}|^2 \int_{E-\epsilon}^{E+\epsilon} \mathrm{sinc}^2\left(\frac{E'/\hbar - \omega}{2}\,t\right)\rho(E')\,dE'$$

where $E \pm \epsilon = \hbar(\omega_2 - \omega_1) \pm \epsilon$ and ρ is density of states.

Use the approximation $\rho(E') \approx \rho(\hbar\omega)$ to obtain

$$P_{tot} = \frac{t^2}{\hbar^2}|M_{21}|^2\rho(\hbar\omega) \int_{E-\epsilon}^{E+\epsilon} \mathrm{sinc}^2\left(\frac{E'/\hbar - \omega}{2}\,t\right)\,dE'$$

Change variable to y where

$$y = \frac{E'/\hbar - \omega}{2}\,t$$

It follows that

$$E' = \frac{2\hbar y}{t} + \hbar\omega$$

and

$$dE' = \frac{2\hbar}{t}\,dy$$

The integration limits transform as

$$E \pm \epsilon \rightarrow \frac{(E \pm \epsilon)/\hbar - \omega}{2}\,t = \frac{Et}{2\hbar} - \frac{\omega t}{2} \pm \frac{\epsilon t}{2\hbar} = \pm\frac{\epsilon t}{2\hbar}$$

Hence

$$P_{tot} = \frac{2t}{\hbar}|M_{21}|^2\rho(\hbar\omega) \int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \mathrm{sinc}^2\,y\,dy$$

Use the approximation

$$\int_{-\epsilon t/2\hbar}^{\epsilon t/2\hbar} \mathrm{sinc}^2\,y\,dy \approx \int_{-\infty}^{\infty} \mathrm{sinc}^2\,y\,dy = \pi$$

to obtain

$$P_{tot} = \frac{2\pi t}{\hbar}|M_{21}|^2\rho(\hbar\omega)$$

The transition rate is the derivative of P_{tot}.

$$R_{1\to2} = \frac{d}{dt}P_{tot} = \frac{2\pi}{\hbar}|M_{21}|^2\rho(\hbar\omega) \tag{4}$$

§3

We started with

$$H_1(\mathbf{r},t) = 2V(\mathbf{r})\cos(\omega t + \phi)$$

to obtain Fermi's golden rule (4). Consequently for

$$H_1(\mathbf{r},t) = V(\mathbf{r})\cos(\omega t + \phi)$$

we would have

$$R_{1\to 2} = \frac{2\pi}{\hbar}\left|\frac{M_{21}}{2}\right|^2 \rho(\hbar\omega)$$